

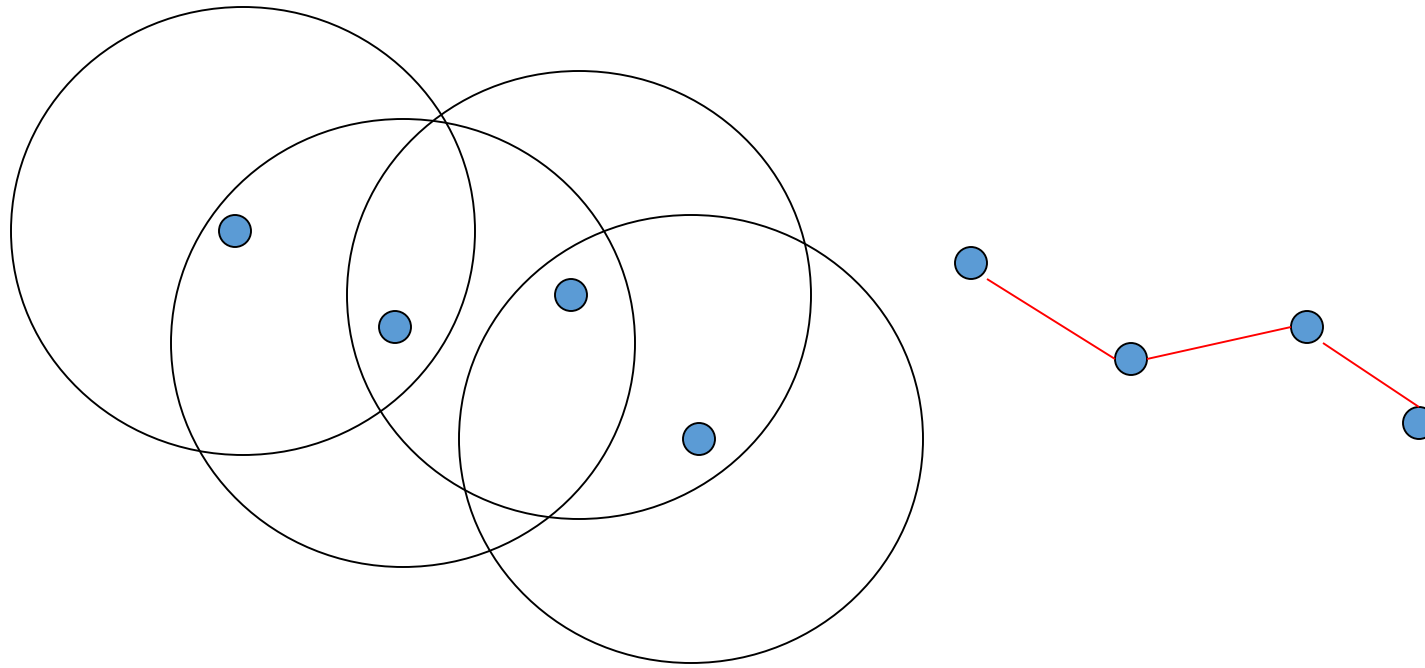
Multihop wireless networks: Routing

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Multi-Hop Wireless

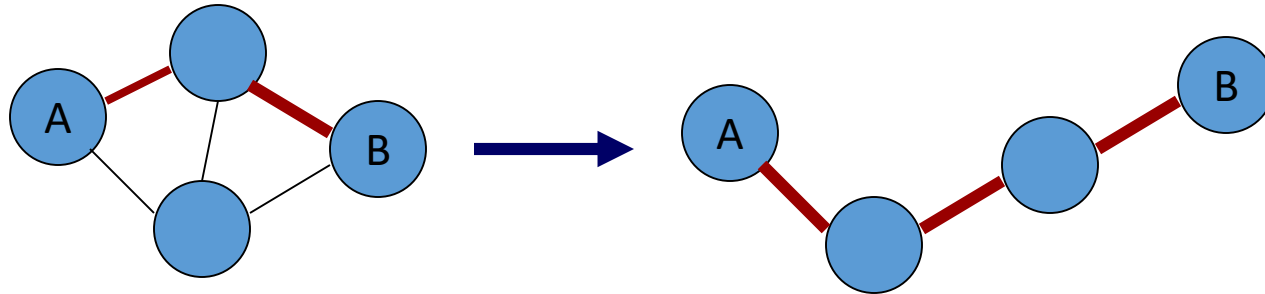
- May need to traverse multiple links to reach destination



- Mobility causes route changes

Mobile Ad Hoc Networks (MANET)

- Host movement frequent
- Topology change frequent



- No cellular infrastructure. Multi-hop wireless links.
- Data must be routed via intermediate nodes.

Why Ad Hoc Networks ?

- Setting up of fixed access points and backbone infrastructure is not always viable
 - Infrastructure may not be present in a disaster area or war zone
 - Infrastructure may not be practical for short-range radios; Bluetooth (range ~ 10m)
- Ad hoc networks:
 - Do not need backbone infrastructure support
 - Are easy to deploy
 - Useful when infrastructure is absent, destroyed or impractical

Many Applications

- **Personal area networking**
 - cell phone, laptop, ear phone, wrist watch
- **Military environments**
 - soldiers, tanks, planes
- **Civilian environments**
 - taxi cab network
 - meeting rooms
 - sports stadiums
 - boats, small aircraft
- **Emergency operations**
 - search-and-rescue
 - policing and fire fighting

Challenges in Mobile Environments

- **Limitations of the Wireless Network**
 - packet loss due to transmission errors
 - variable capacity links
 - frequent disconnections/partitions
 - limited communication bandwidth
 - Broadcast nature of the communications
- **Limitations Imposed by Mobility**
 - dynamically changing topologies/routes
 - lack of mobility awareness by system/applications
- **Limitations of the Mobile Computer**
 - short battery lifetime
 - limited capacities

Effect of mobility on the protocol stack

- **Application**
 - new applications and adaptations
- **Transport**
 - congestion and flow control
- **Network**
 - addressing and routing
- **Link**
 - media access and handoff
- **Physical**
 - transmission errors and interference

Routing Protocol

- Figure out a path from source to destination.
- Basic techniques of routing over multiple hops (wired or wireless):
 - **Link-state (LS) routing**: propagate information about your neighbors to everyone in the network. With the complete network topology, every node computes shortest paths. Lots of overhead in flooding information, but no convergence issues. The default protocol used for intra-domain routing (e.g., OSPF) in wired networks.
 - **Distance vector (DV) routing**: propagate information about all destinations you know (including yourself) to your neighbors. Suffers from the “count to infinity” problem in the basic version. Split horizon and other techniques fix the count to infinity problem. A variant of distance vector called path vector is used for inter-domain routing (BGP) in wired networks.

Routing and Mobility

- Finding a path from a source to a destination
- Issues
 - Frequent route changes
 - amount of data transferred between route changes may be much smaller than traditional networks
 - Route changes may be related to host movement
 - Low bandwidth links
- Goal of routing protocols
 - decrease routing-related overhead
 - find short routes
 - find “stable” routes (despite mobility)

Routing Protocols

- **Proactive protocols**
 - Traditional distributed shortest-path protocols
 - Maintain routes between every host pair at all times
 - Based on periodic updates; High routing overhead
 - Distance vector based (e.g., DSDV (destination sequenced distance vector))
- **Reactive protocols**
 - Determine route if and when needed
 - Source initiates route discovery; (e.g., DSR(dynamic source routing))
- Which approach achieves a better trade-off depends on the traffic and mobility patterns

Basic terminology

- A route has 3 main components:
 - The **destination** to which it leads to
 - The **next hop** along the path to the destination, to which we must forward data
 - A **metric** (e.g., hop count) that indicates the desirability of the route
- A routing table is a list of routes to all possible destinations.
- A forwarding table is a summary of the routing table by considering the best routes to be used for forwarding.

DSDV

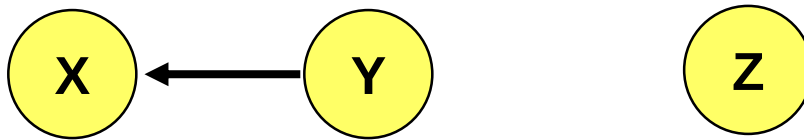
- Classic DV suffers from **count-to-infinity** problem. The standard fix is split horizon routing, where you do not announce a route advertised by router X back on the link to X. **However, wireless broadcast medium has no notion of a link, so split horizon does not work.**
- Fix in DSDV: use sequence numbers to denote freshness of routes. **Destinations update sequence numbers if some major change in their state has occurred.**
- Update your routing table entry only if:
 - A route of a higher sequence number appears OR
 - A route of the same sequence but better metric appears

Destination-Sequenced Distance-Vector (DSDV)

- Each node maintains a routing table which stores
 - next hop, cost metric towards each destination
 - a sequence number that is created by the destination itself
- Each node periodically forwards routing table to neighbors
 - Each node increments and appends its sequence number when sending its local routing table
- Each route is tagged with a sequence number; routes with greater sequence numbers are preferred
- Each node advertises a monotonically increasing even sequence number for itself
- When a node decides that a route is broken, it increments the sequence number of the route and advertises it with infinite metric
- Destination advertises new sequence number

Destination-Sequenced Distance-Vector (DSDV)

- When X receives information from Y about a route to Z
 - Let destination sequence number for Z at X be $S(X)$, $S(Y)$ is sent from Y



- If $S(X) > S(Y)$, then X ignores the routing information received from Y
- If $S(X) = S(Y)$, and cost of going through Y is smaller than the route known to X, then X sets Y as the next hop to Z
- If $S(X) < S(Y)$, then X sets Y as the next hop to Z, and $S(X)$ is updated to equal $S(Y)$

DSDV (2)

- Periodically, every node broadcasts for each destination (including itself):
 - Destination address
 - Sequence number
 - Metric
- Routing updates are also triggered by significant events such as updated best routes.
- Pro: finds good paths in a loop-free manner
- Con: Lots of overhead in terms of periodic routing broadcasts.

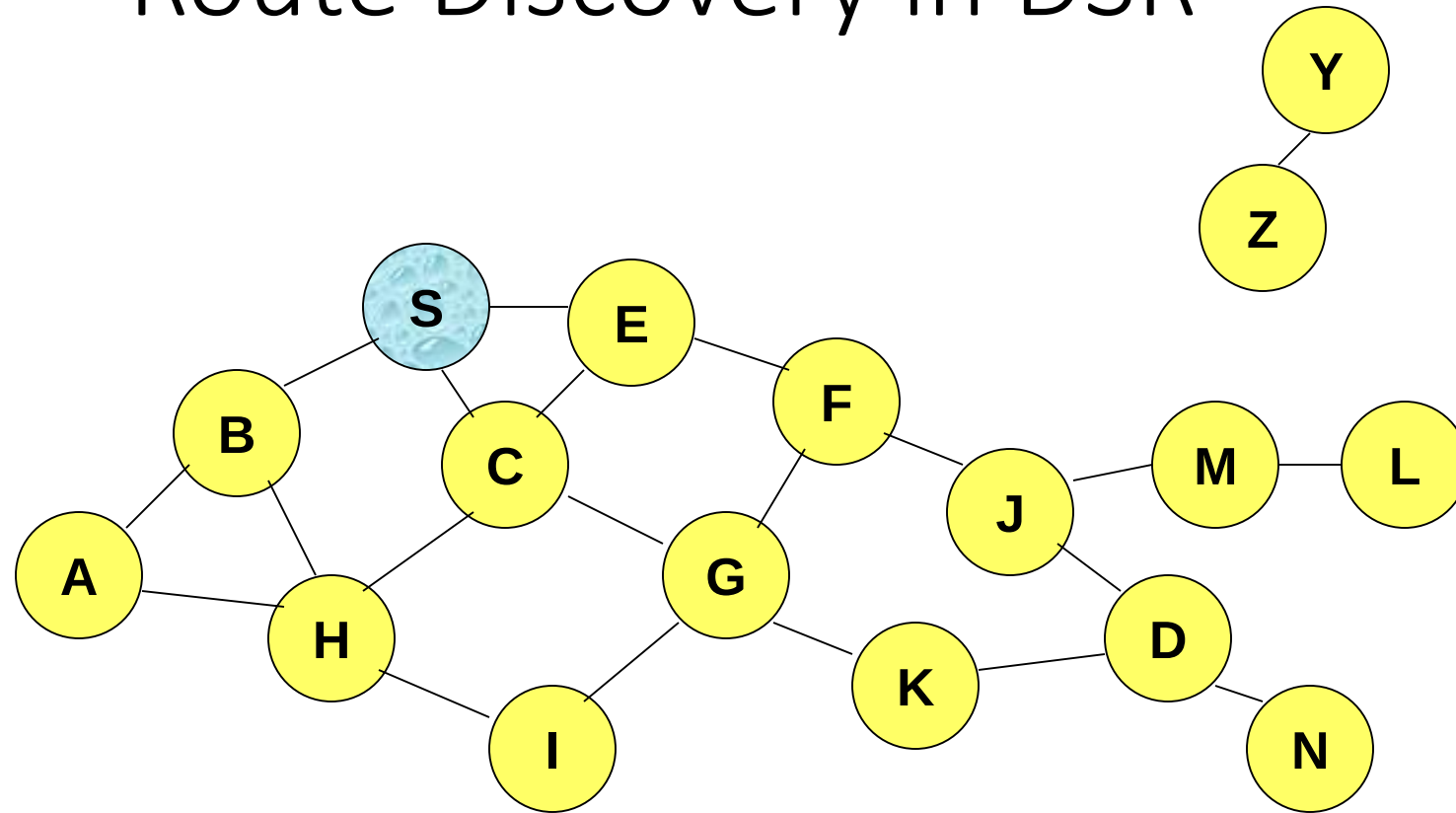
DSR

- Source routing: the source discovers the route and places the complete path in each packet. Intermediate routers only need to follow the path.
- DSR has two steps: route discovery and route maintenance.
- Route discovery: when a source S has to send a packet to destination D, it broadcasts a route request (RREQ) packet to all its neighbors. Every RREQ has a unique id.
 - Every neighbor appends itself to the path, increments metric, and forwards the route request.
 - A node does not forward a route request if it has already seen it.
 - When the route request reaches destination D, it picks the best path, and sends this path back to the source in a route reply (RREP).

Dynamic Source Routing (DSR)

- When node S wants to send a packet to node D, but does not know a route to D, node S initiates a **route discovery**
- Source node S floods **Route Request (RREQ)**
- Each node *appends own identifier* when forwarding RREQ

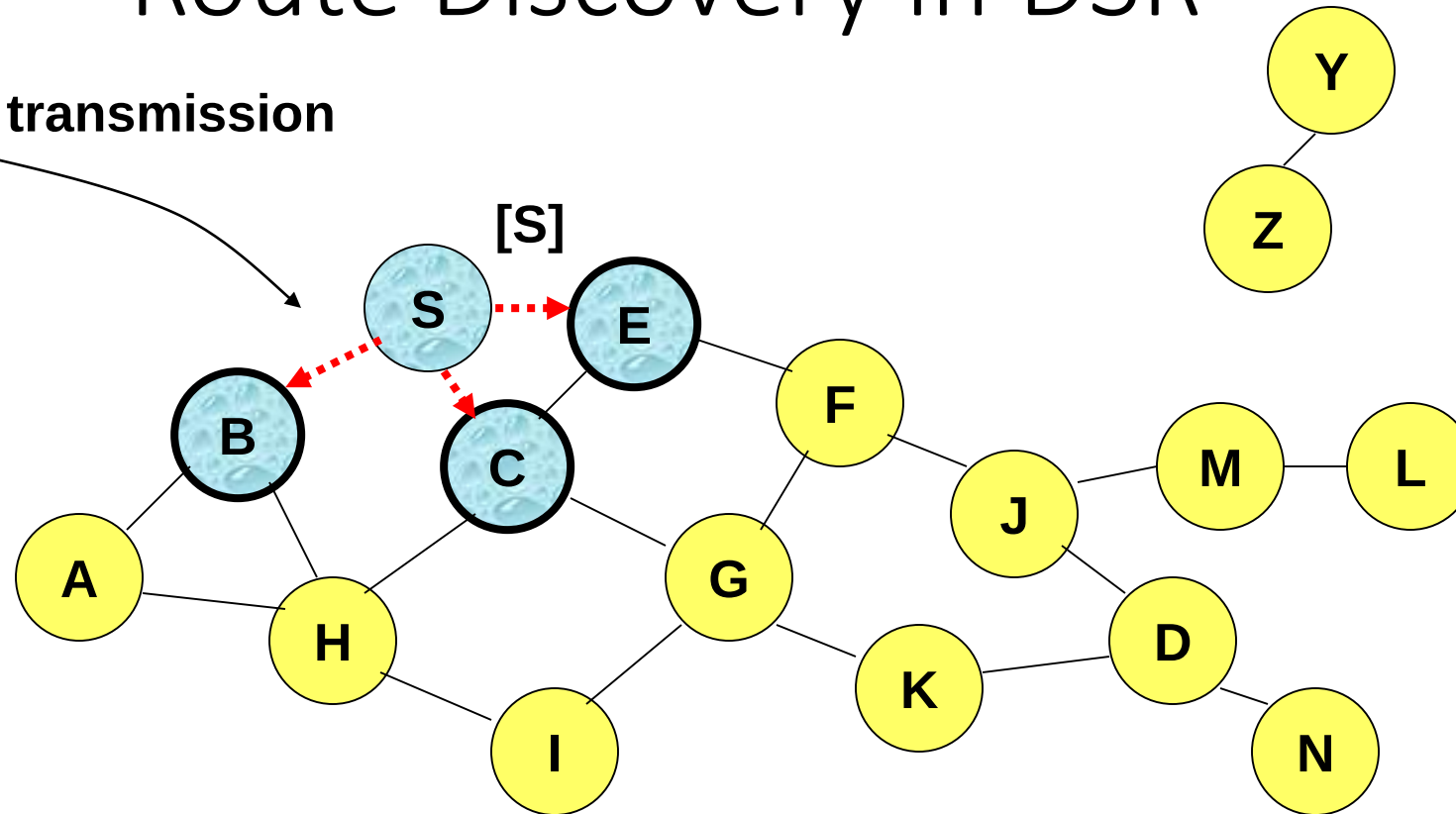
Route Discovery in DSR



Represents a node that has received RREQ for D from S

Route Discovery in DSR

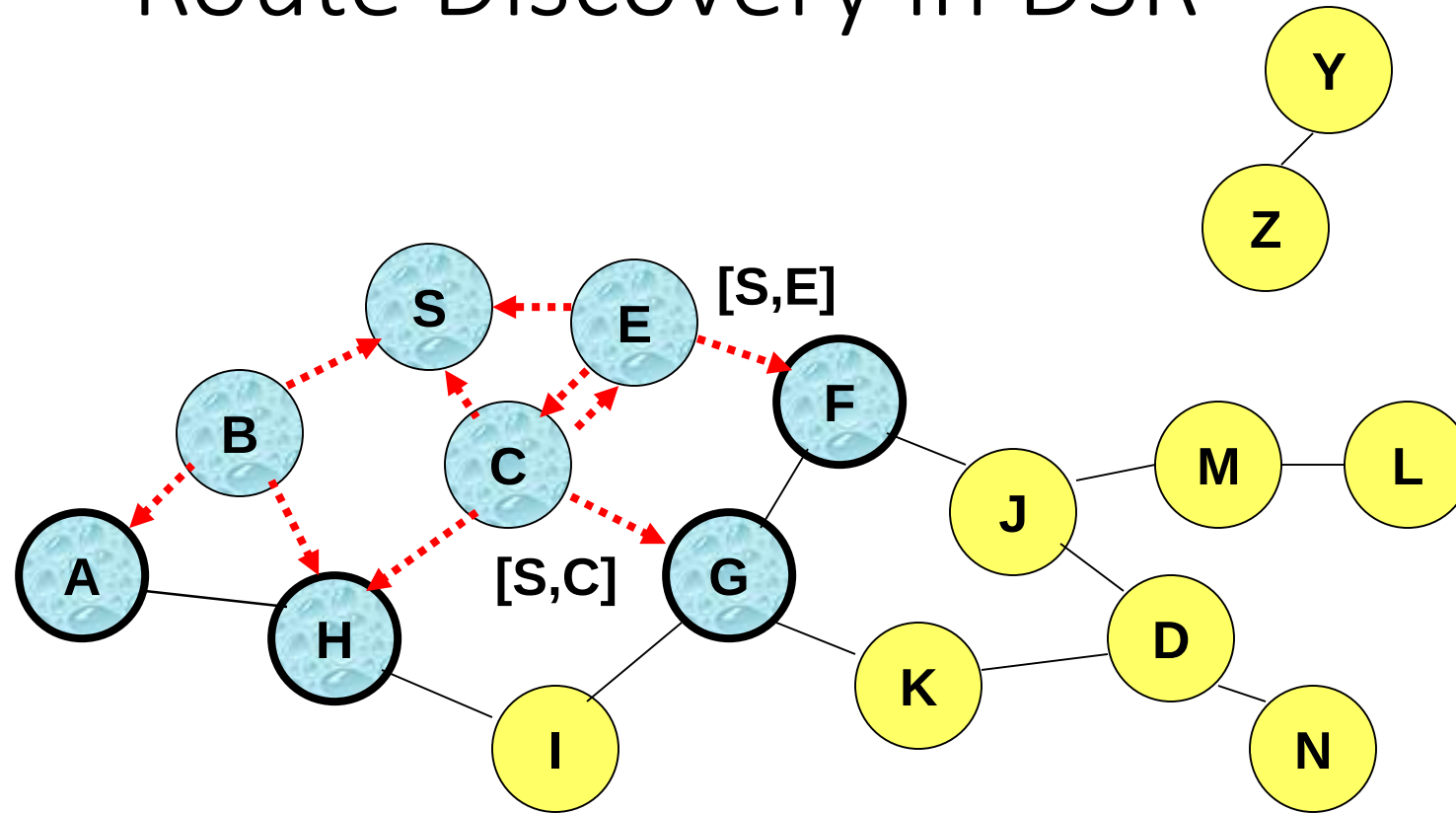
Broadcast transmission



.....→ Represents transmission of RREQ

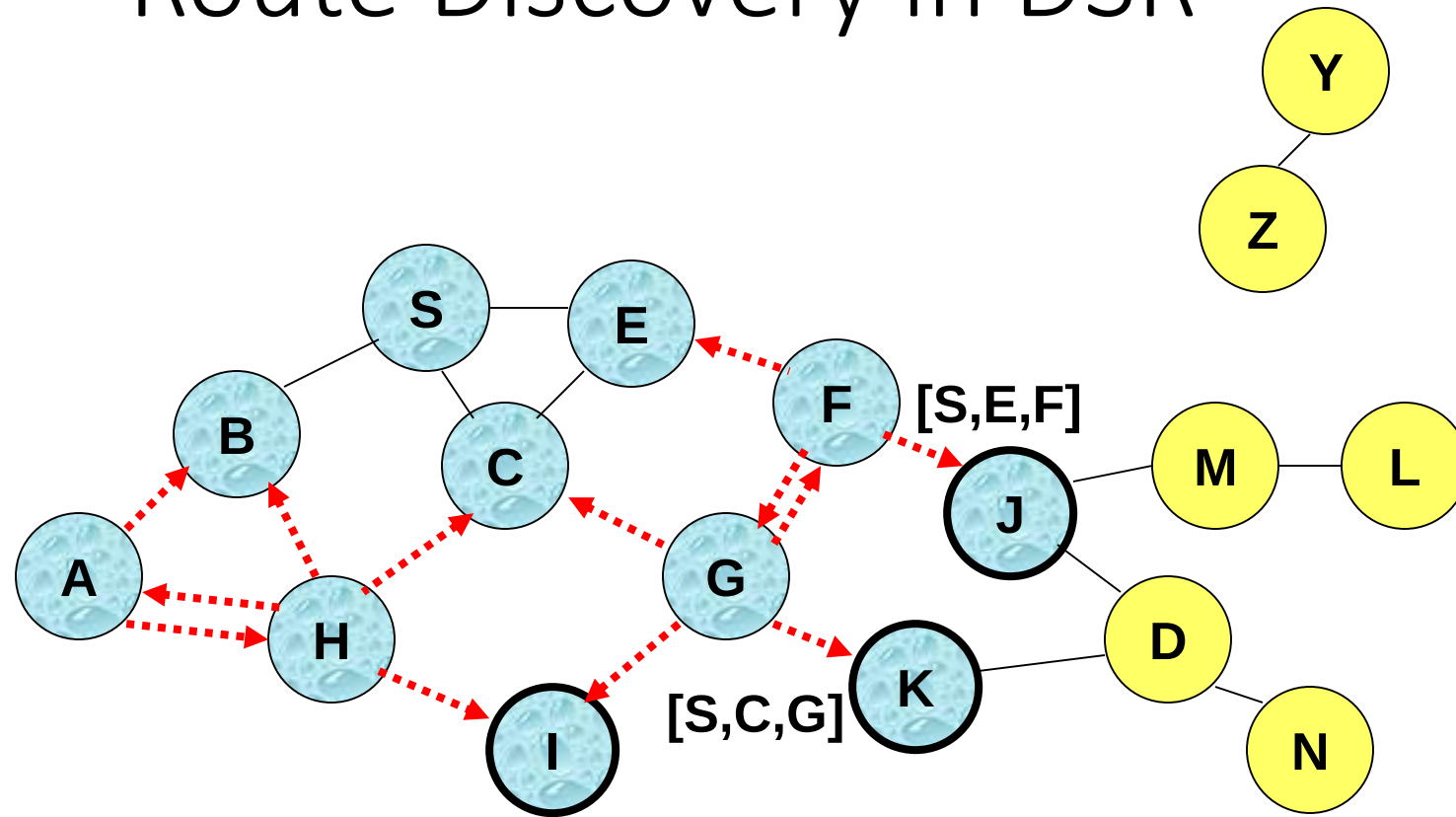
[X,Y] Represents list of identifiers appended to RREQ

Route Discovery in DSR



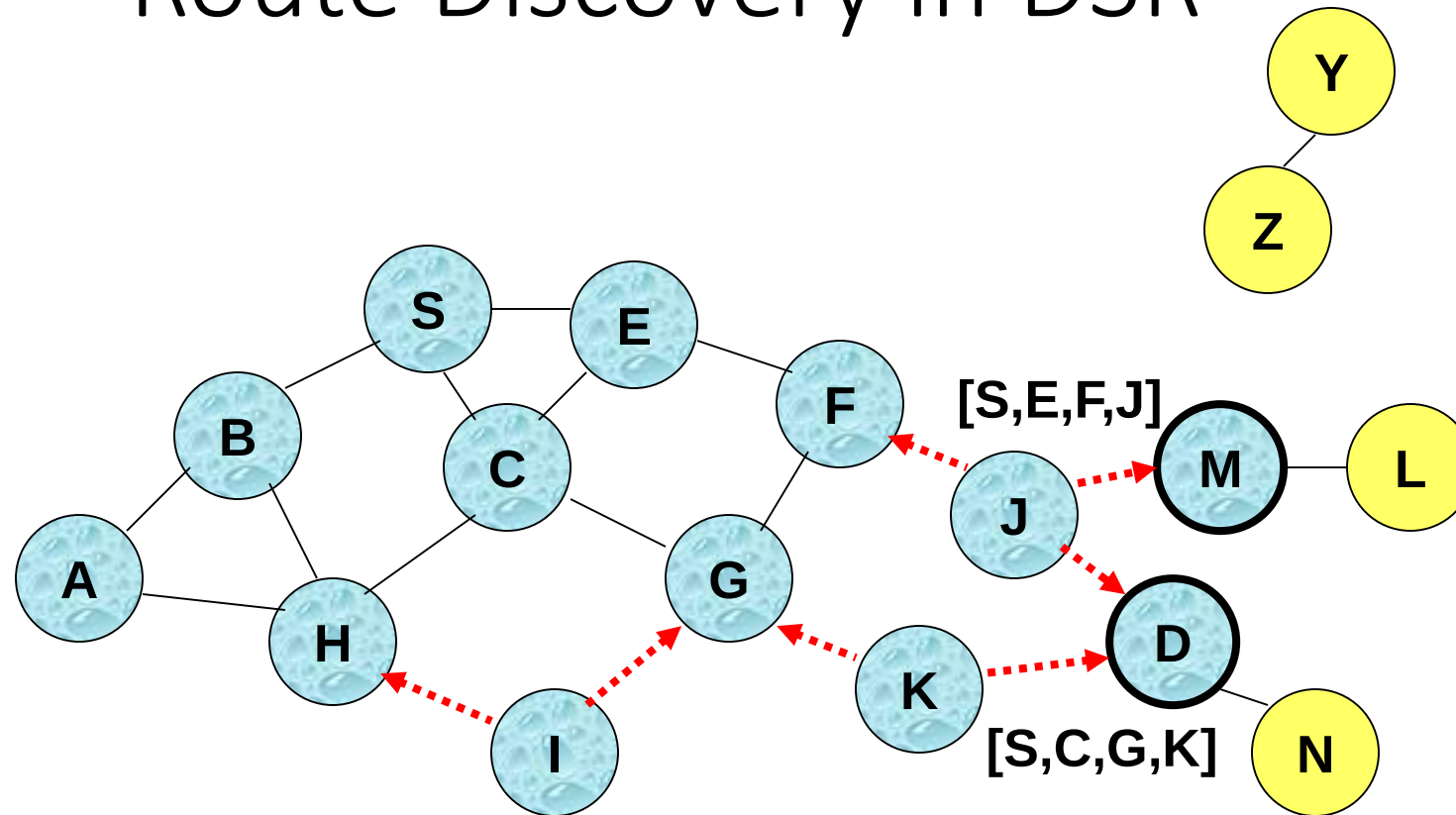
- Node H receives packet RREQ from two neighbors:
potential for collision

Route Discovery in DSR



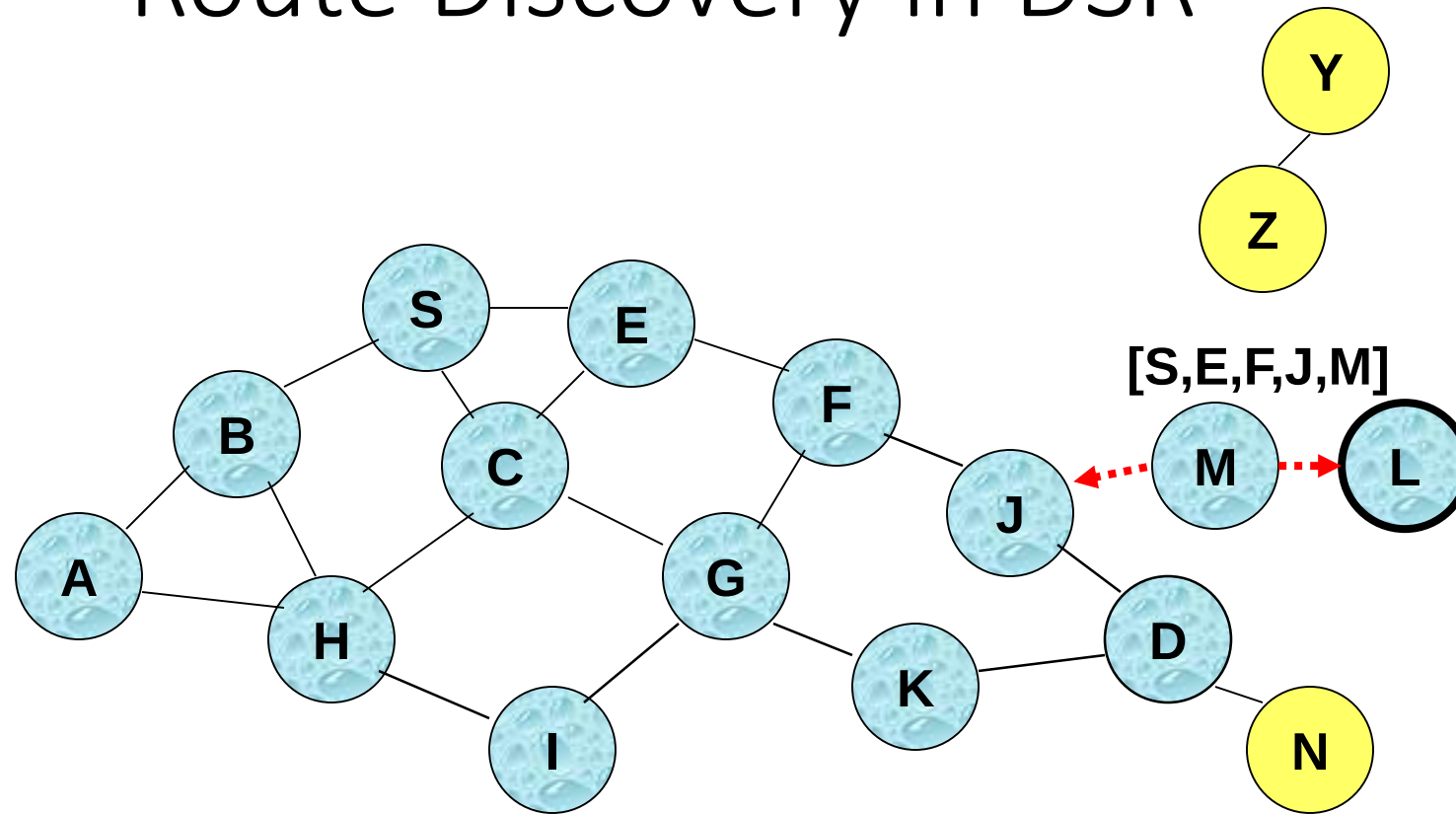
- Node C receives RREQ from G and H, but does not forward it again, because node C has **already forwarded RREQ** once

Route Discovery in DSR



- Nodes J and K both broadcast RREQ to node D
- Since nodes J and K are **hidden** from each other, their **transmissions may collide**

Route Discovery in DSR

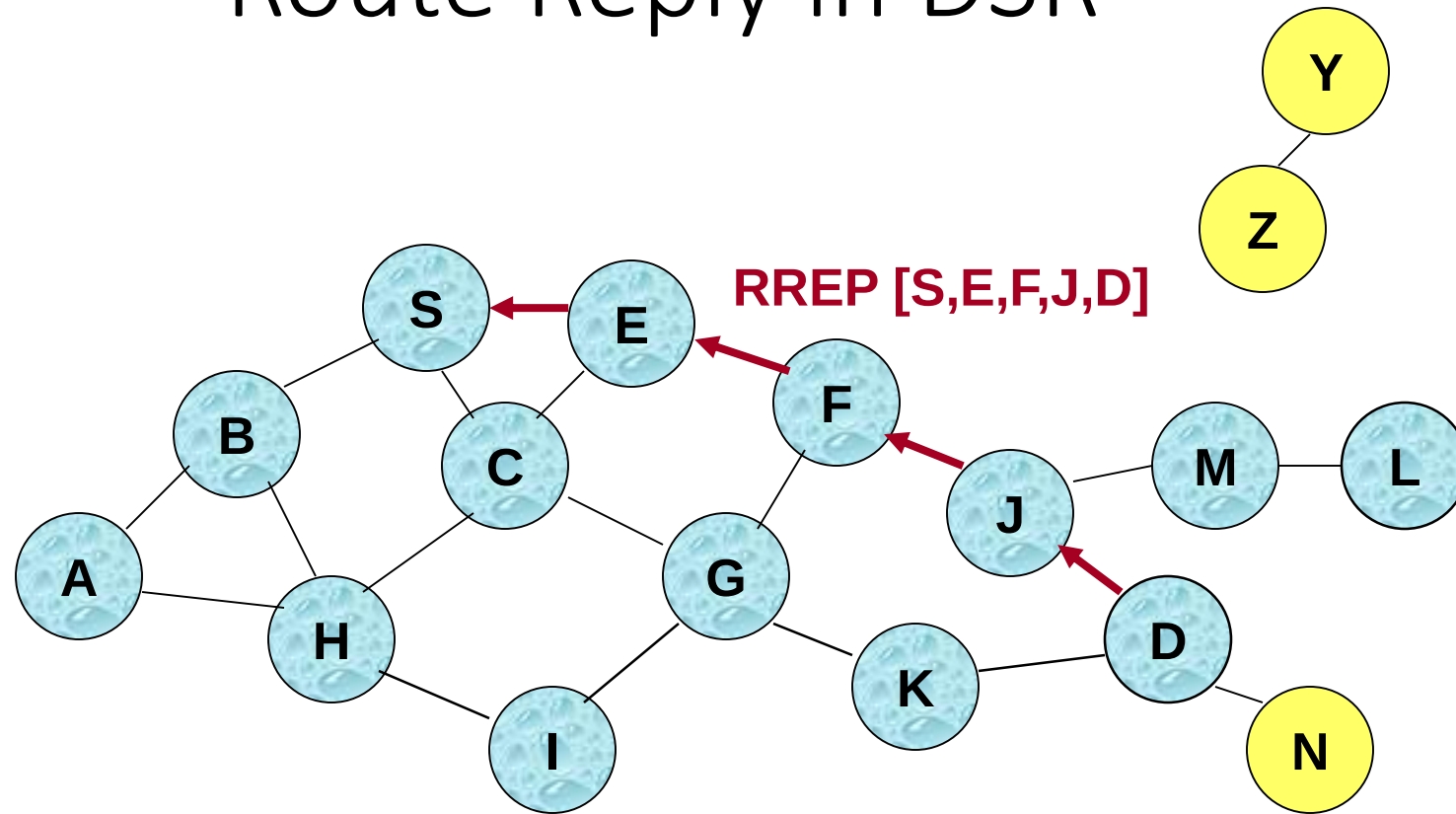


- Node D **does not forward** RREQ, because node D is the **intended target** of the route discovery

Route Discovery in DSR

- Destination D on receiving the first RREQ, sends a **Route Reply (RREP)**
- RREP is sent on a route obtained by **reversing** the route appended to received RREQ
- RREP **includes the route** from S to D on which RREQ was received by node D

Route Reply in DSR

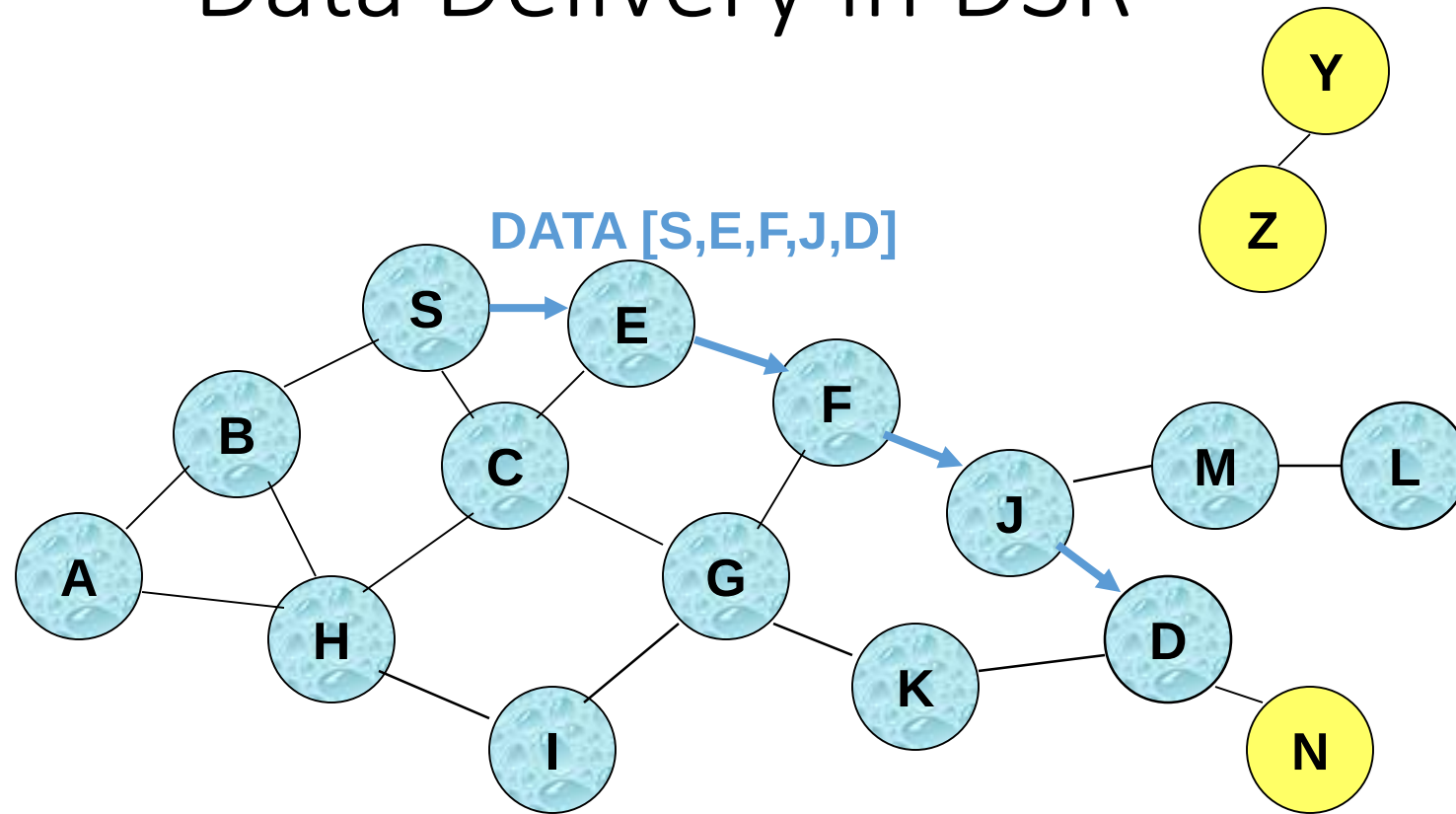


← Represents RREP control message

Dynamic Source Routing (DSR)

- Node S on receiving RREP, caches the route included in the RREP
- When node S sends a data packet to D, the entire route is included in the packet header
 - hence the name **source routing**
- Intermediate nodes use the **source route** included in a packet to determine to whom a packet should be forwarded

Data Delivery in DSR



Packet header size grows with route length

DSR (2)

- Route maintenance: if any node notices large losses while using a route, it informs the source, which starts a new route discovery with the next sequence number.
- Pro:
 - Every packet has the complete path, so intermediate routers keep no state.
 - Route discovery is only initiated when there is a need, and only at the routers that are likely to be in the path
- Con:
 - Extra overhead in each packet.
 - First packet has a large delay due to the overhead of route discovery.

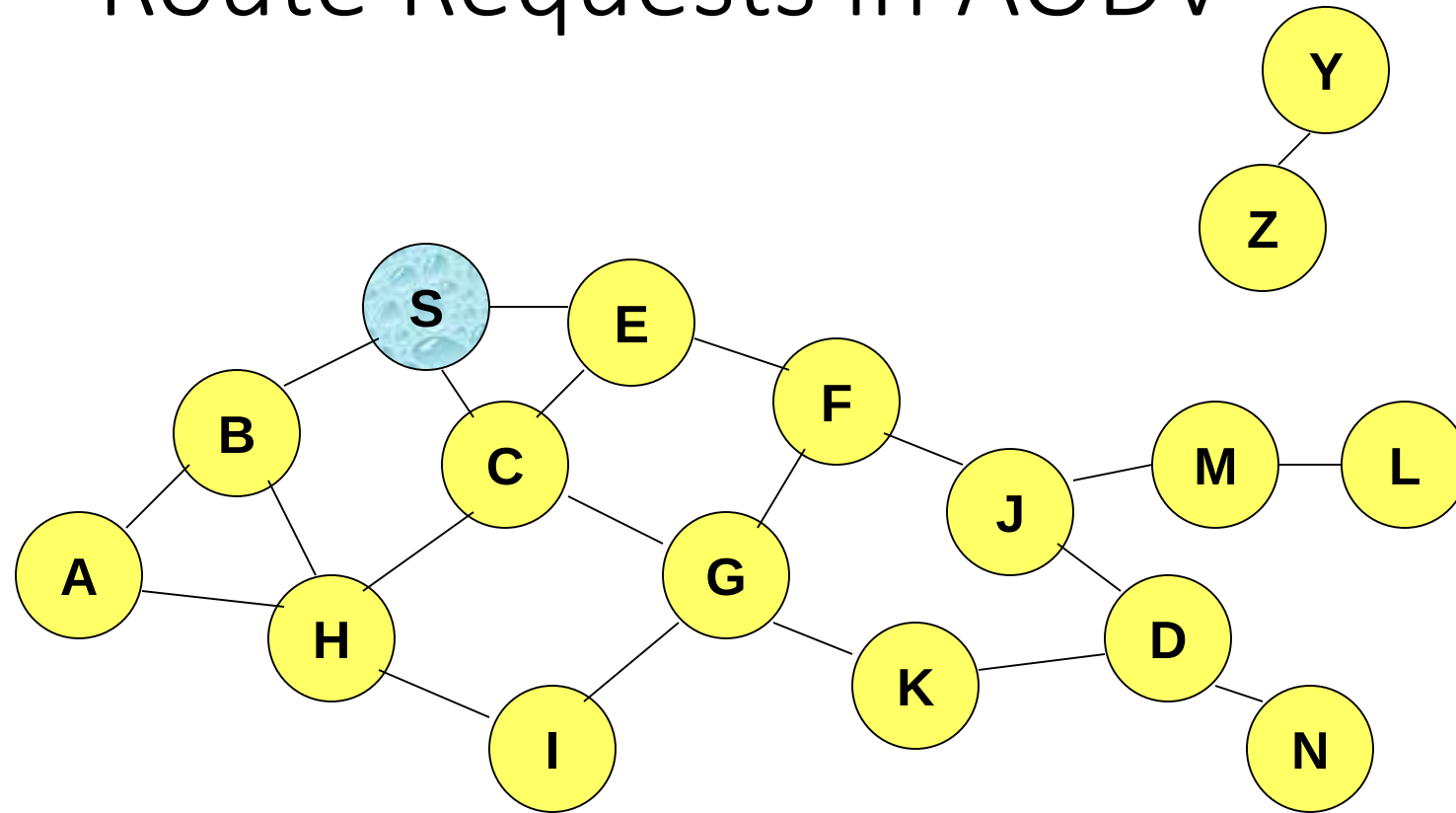
AODV

- Combines the best of DSDV and DSR.
- Route discovery only when needed (RREQ, RREP)
- However, distance vector type routing table entries are setup during route discovery.
 - Every node that receives the RREQ from source sets up reverse path routing table entries pointing to the source.
 - Similarly, every node that receives RREP sets up routing table entries pointing to the destination on the forward path.
- That is, RREQ and RREP also serve as route advertisements in DV.
- There is also a sequence number associated with source and destination corresponding to the seq no. in DSDV.

AODV

- **Route Requests (RREQ)** are forwarded in a manner similar to DSR
- When a node re-broadcasts a Route Request, it sets up a reverse path pointing towards the source
 - AODV assumes symmetric (bi-directional) links
- When the intended destination receives a Route Request, it replies by sending a **Route Reply (RREP)**
- Route Reply travels along the reverse path set-up when Route Request is forwarded

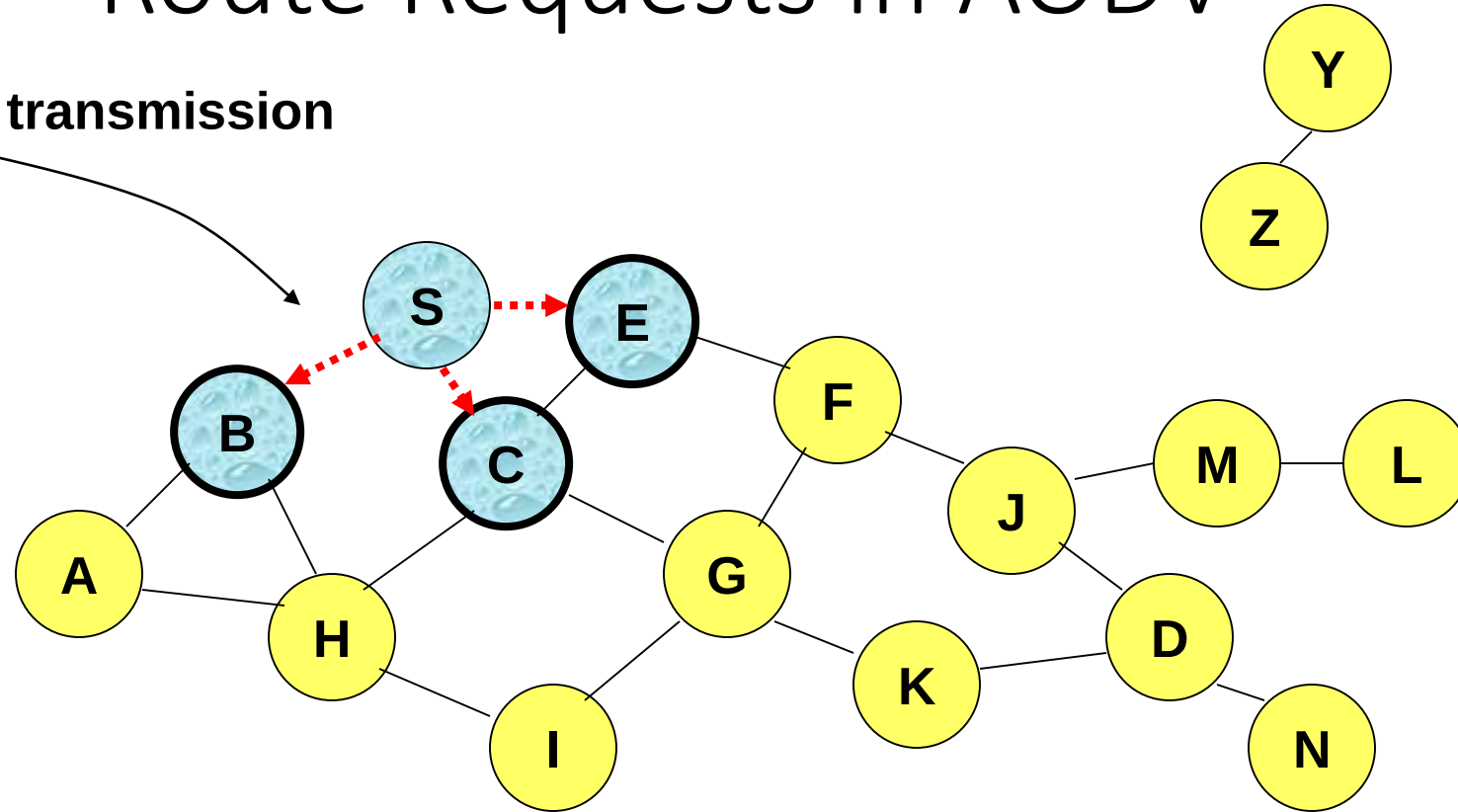
Route Requests in AODV



Represents a node that has received RREQ for D from S

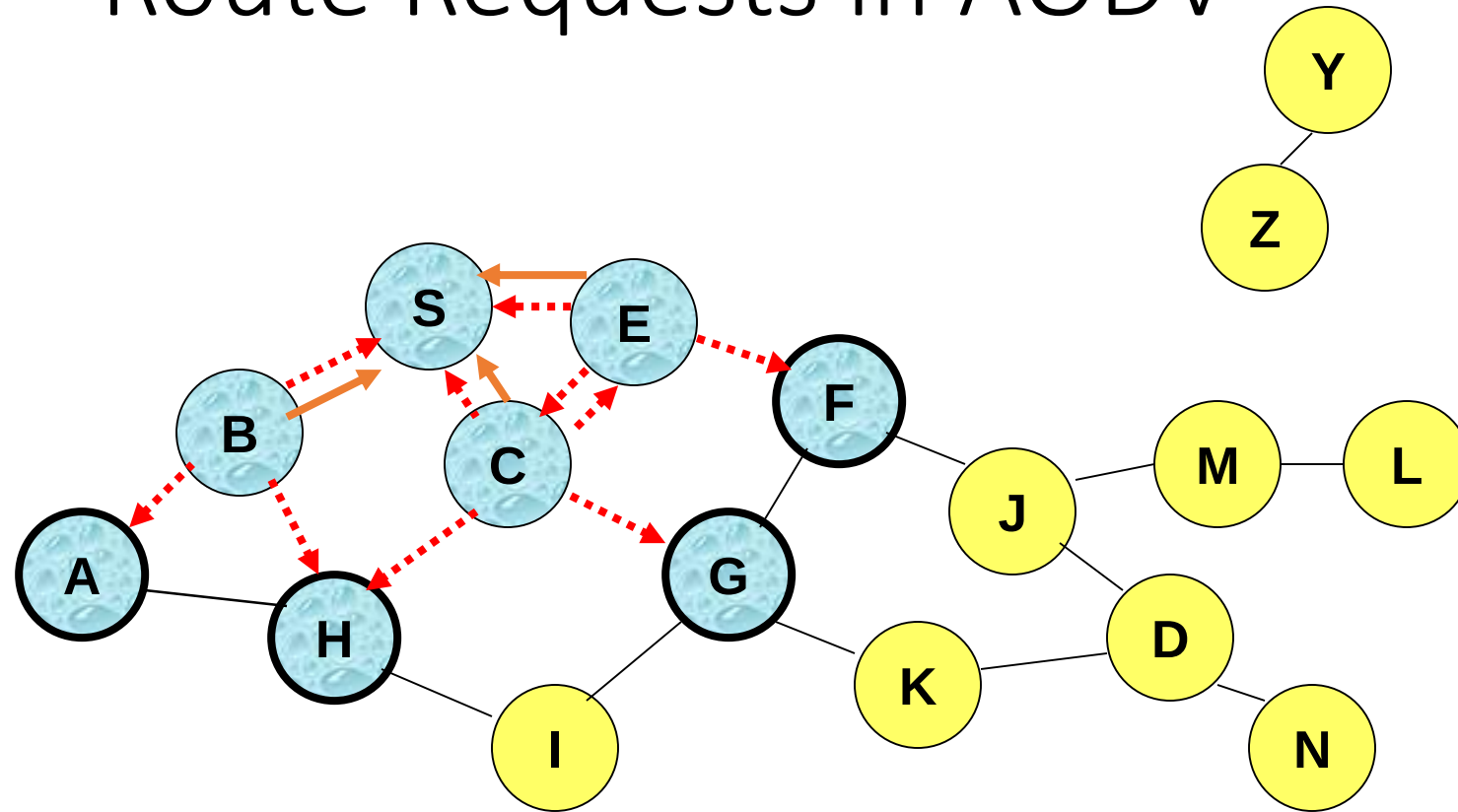
Route Requests in AODV

Broadcast transmission



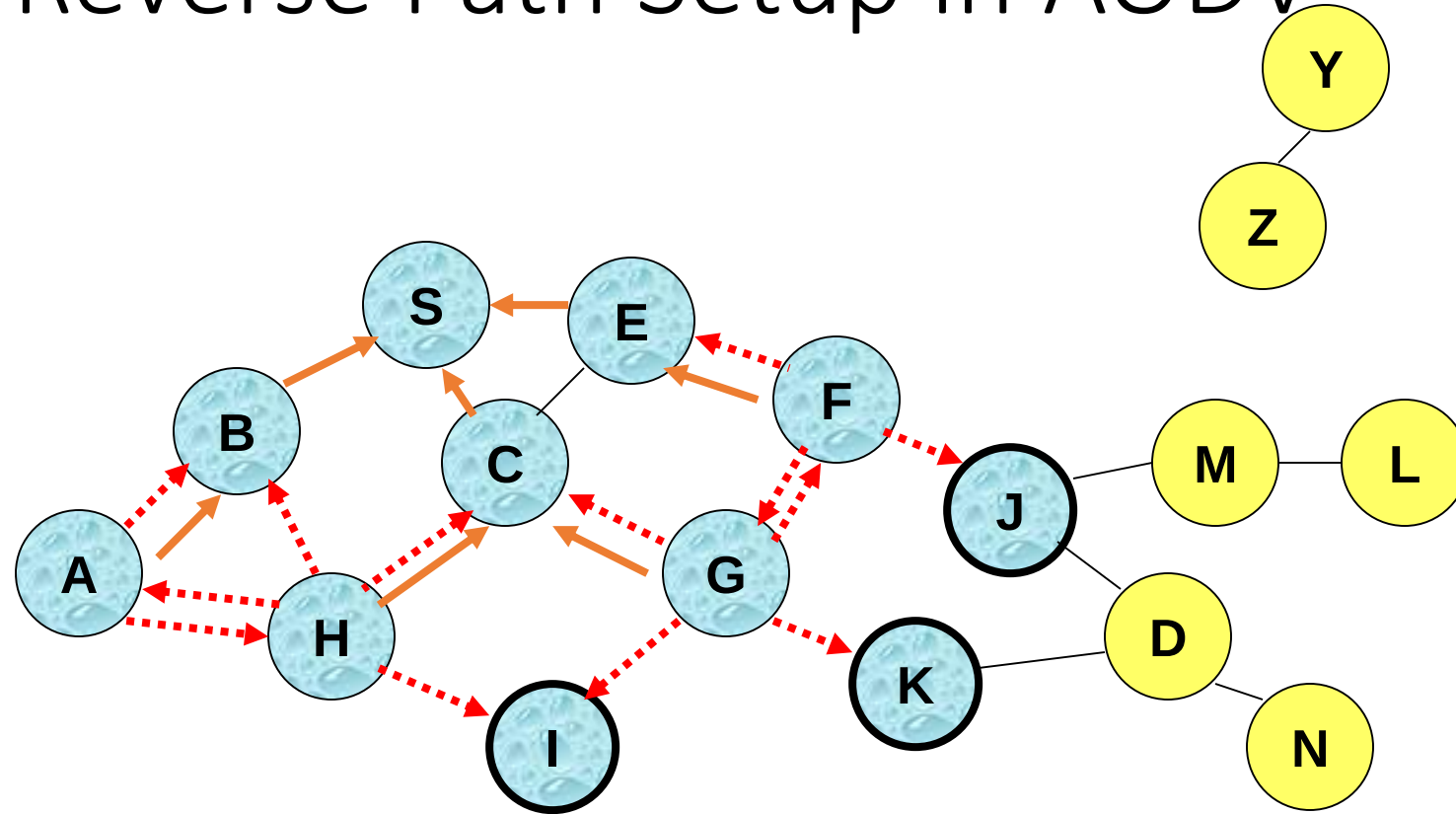
.....> Represents transmission of RREQ

Route Requests in AODV



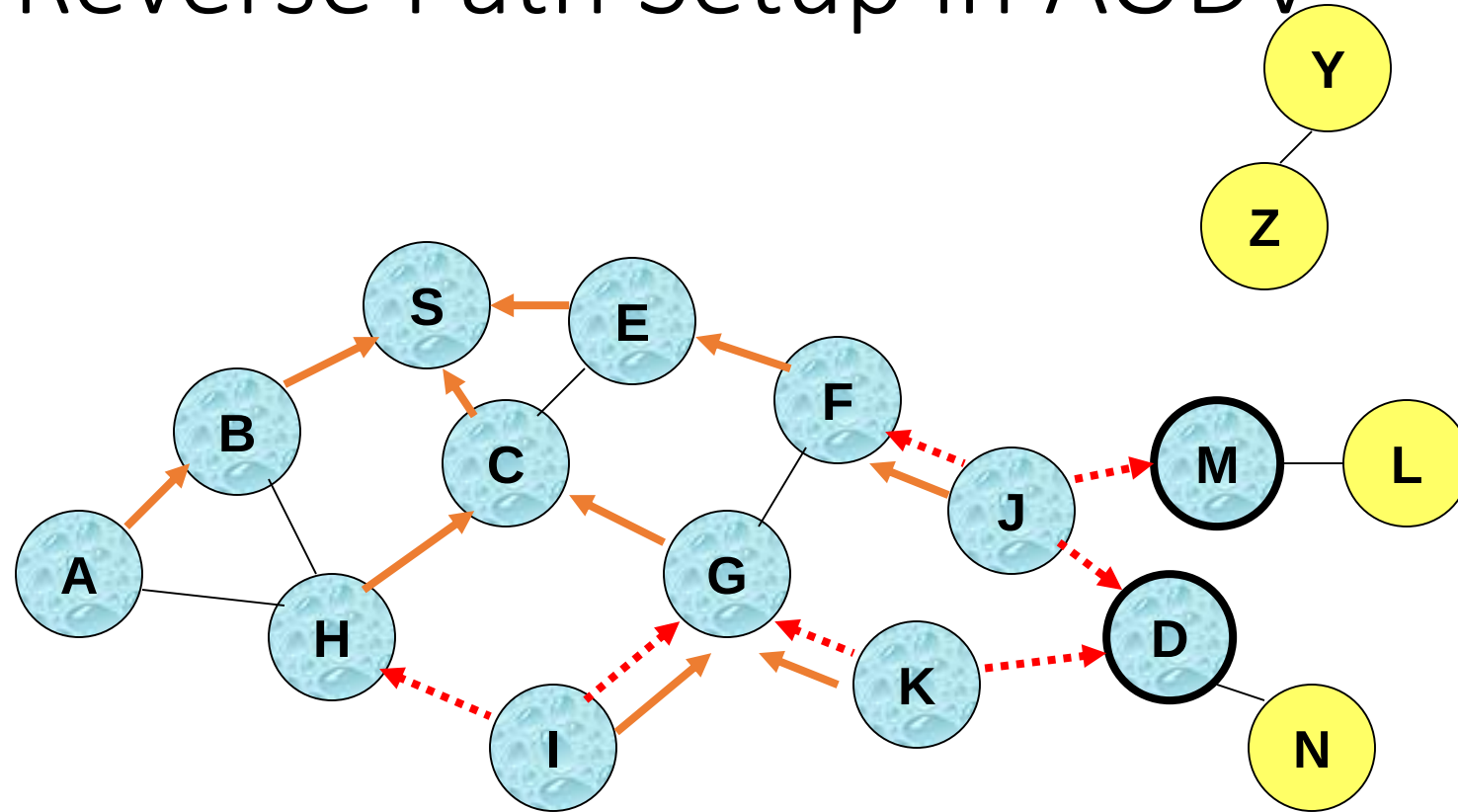
← Represents links on Reverse Path

Reverse Path Setup in AODV

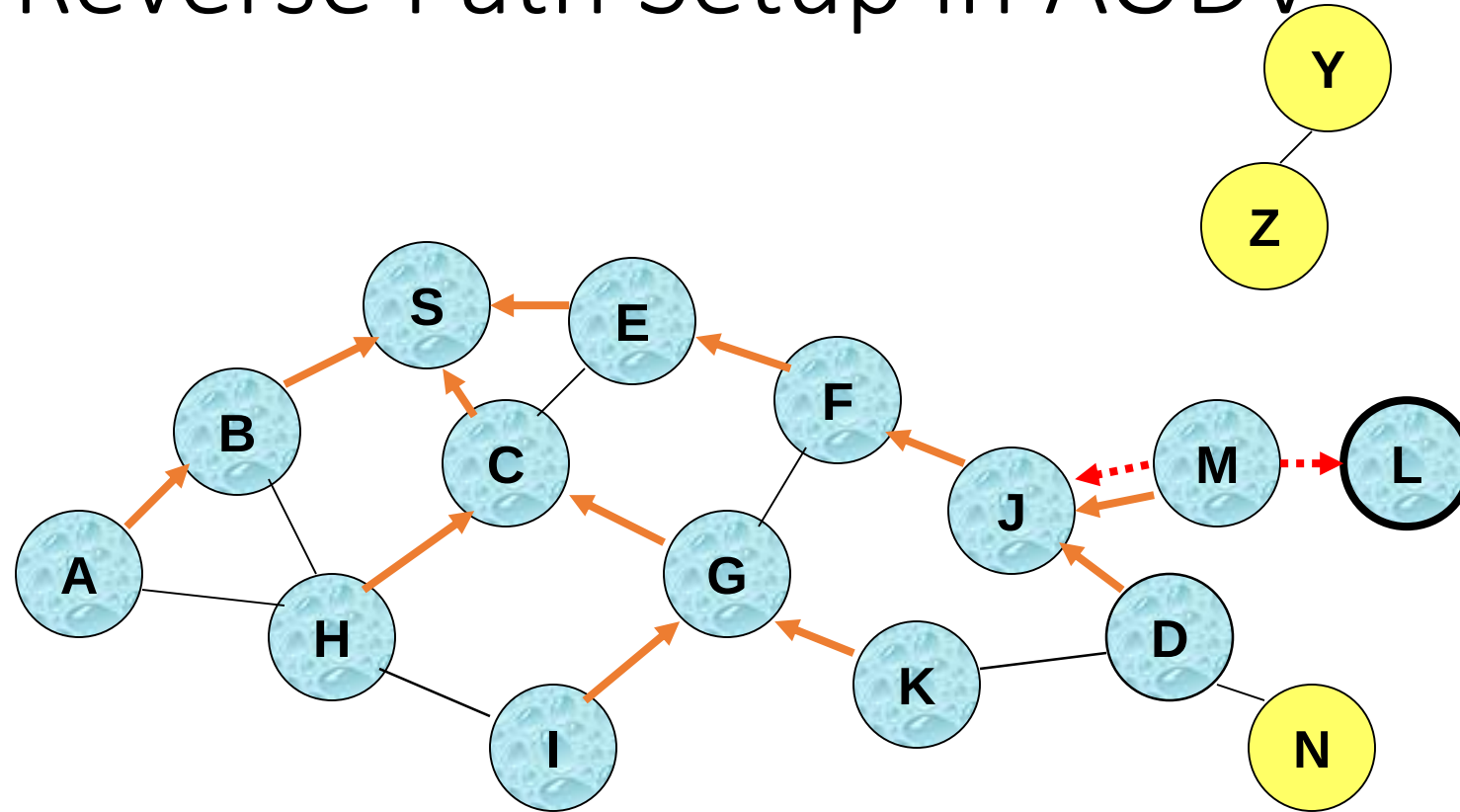


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Reverse Path Setup in AODV

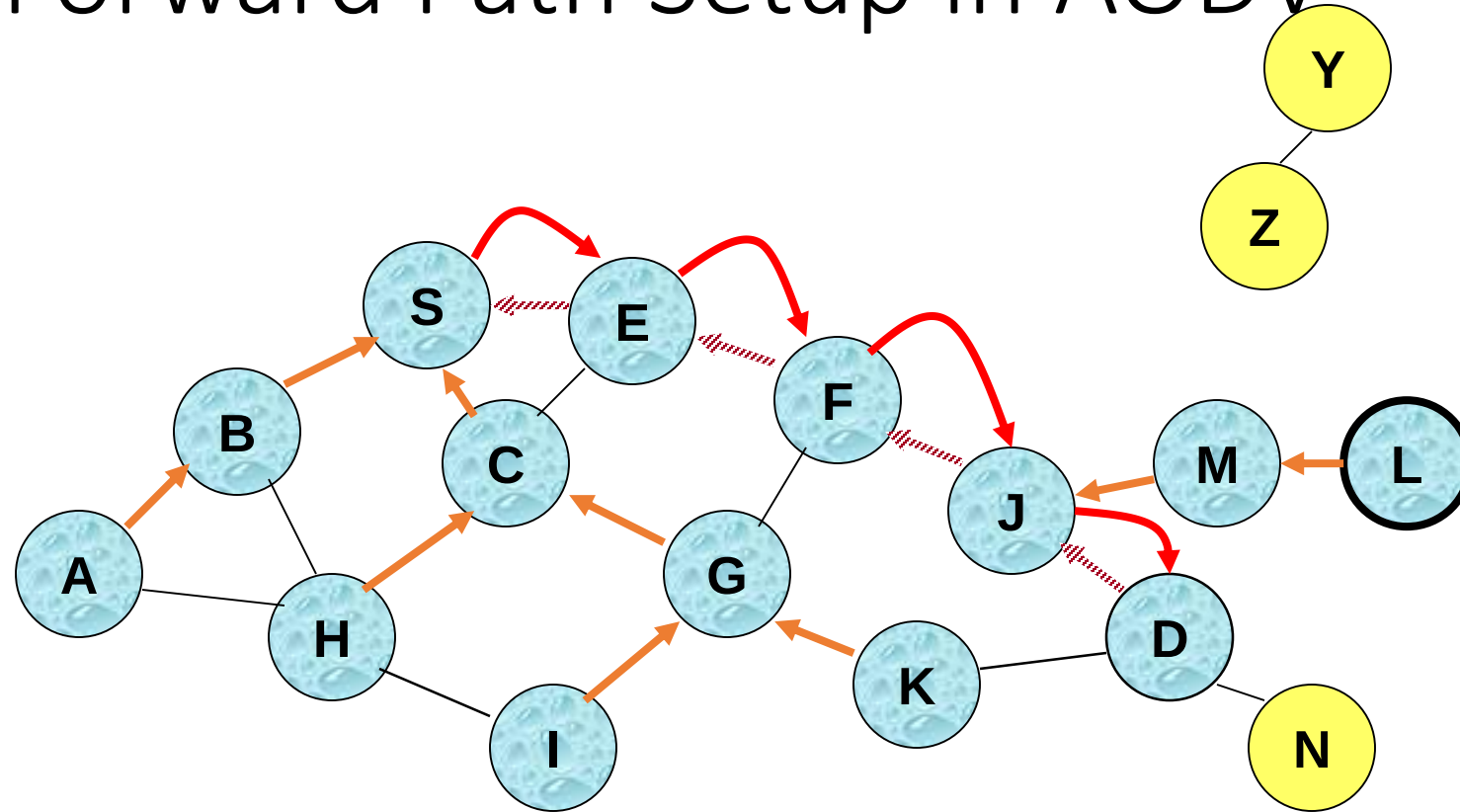


Reverse Path Setup in AODV



- Node D **does not forward** RREQ, because node D is the **intended target** of the RREQ

Forward Path Setup in AODV



Forward links are setup when RREP travels along the reverse path



Represents a link on the forward path

Route Request and Route Reply

- Route Request (RREQ) includes the last known **sequence number** for the destination
- An intermediate node may also send a Route Reply (RREP) provided that it knows a **more recent path** than the one previously known to sender
- Intermediate nodes that forward the RREP, also record the next hop to destination
- A routing table entry maintaining a **reverse path** is purged after a timeout interval
- A routing table entry maintaining a **forward path** is purged if *not used* for a ***active_route_timeout*** interval

Link Failure

- A neighbor of node X is considered **active** for a routing table entry if the neighbor sent a packet within ***active_route_timeout*** interval which was forwarded using that entry
- Neighboring nodes periodically exchange **hello** message
- When the next hop link in a routing table entry breaks, all **active** neighbors are informed
- Link failures are propagated by means of **Route Error (RERR)** messages, which also update destination sequence numbers

Route Error

- When node X is unable to forward packet P (from node S to node D) on link (X,Y), it generates a RERR message
- Node X increments the destination sequence number for D cached at node X
- The incremented sequence number N is included in the RERR
- When node S receives the RERR, it initiates a new route discovery for D using destination sequence number at least as large as N
- When node D receives the route request with destination sequence number N , node D will set its sequence number to N , unless it is already larger than N

AODV: Summary

- Routes need not be included in packet headers
- Nodes maintain routing tables containing entries only for routes that are in active use
- At most one next-hop per destination maintained at each node
 - DSR may maintain several routes for a single destination
- Sequence numbers are used to avoid old/broken routes
- Sequence numbers prevent formation of routing loops
- Unused routes expire even if topology does not change

Multihop routing metrics

- Many possible metrics.
- Simple: HOP count between two nodes. Pick the path with least number of hops.
- More sophisticated metrics: expected number of transmissions ETX (accounting for link-layer losses), or expected transmission time ETT (accounting for different bit rates)
- Load based metrics like RTT are not preferred as they cause oscillations.

Geographic routing

- Another routing technique where you use geographic coordinates to forward traffic.
- All nodes know the geographic coordinates of each other.
- To send a packet to a destination, you send it to a node that is geographically closer than yourself.
- Many proposals in this space. We will not cover these. The goal here is to introduce the basic notion.

Security

- Multihop routing protocols are vulnerable to many security attacks (much like wired routing protocols like BGP)
- For example, a malicious node can falsely announce that it has a short path to a destination, and attract traffic to flow through it (after which it can eavesdrop, alter, or drop the traffic).
- Implementing hop-by-hop authentication (to check validity of next hop) or source authentication (to check a node is who it claims to be) requires cryptographic solutions that depend on knowing cryptographic keys of each other

Attacks on routing protocols

- attack mechanisms
 - eavesdropping, replaying, modifying, and deleting control packets
 - fabricating control packets containing fake routing information (forgery)
 - fabricating control packets under a fake identity (spoofing)
 - dropping data packets (attack against the forwarding function)
 - wormholes and tunneling
 - rushing
- types of attacks
 - route disruption
 - route diversion
 - creation of incorrect routing state
 - generation of extra control traffic
 - creation of a gray hole